



Prudent 5

Disinfectant with 5% Peracetic Acid

Prudent 5 is an oxidizing agent broadly used as a cold sanitizer against microbes like bacteria, fungi, and viruses in food, breweries, wineries, dairy, food processing and pharmaceutical industry and as cooling tower water disinfectant (it effectively prevents bio-film formation and controls Legionella bacteria).

Key Benefits

- Versatile product, can be used in many industries
- Excellent micro-biocidal activity
- Non-corroding & non-foaming at normal dilutions
- More effective than chlorine or quaternary amines in sanitization
- Excellent disinfectant for cold plant operations
- Easily disinfects at cold temperature
- Degradable and eco-friendly (no chlorinated by-product)
- Rinsing after use not required
- Quick results

Application Area

- Dairies, wineries, breweries and beverage plants
- Dairy disinfection (equipment/water disinfection, thermal fogging)
- Antimicrobial disinfection in breweries and wineries, equipment disinfection (tanks, vats, evaporators etc.)
- RO/NF/UF membrane disinfection,
- Processing/packaging equipment surfaces in food industries (particularly sugar industries)
- Pharmaceutical disinfection (medical equipment, cold sterilization of dentures, plastic implants, syringes, thermally sensitive nutritive media, disinfection of hemodialysis systems and decontamination of liquid and solid medical wastes in hospitals)

Application Method

Use at concentrations between 0.04 - 2% w/w (0.03 - 1.8 v/v) depending on application. Always rinse thoroughly after use.

For more details, please contact your local Buzil Rossari representative.



Prudent 5

Disinfectant with 5% Peracetic Acid

Technical Data

Appearance	Bright colorless liquid
pH (1% solution at 20°C)	< 2.0
Odor	Acrid odor
Solubility	High solubility in water
Density at 20°C	1.12 g/ml
Available Oxygen (% w/w)	~13.5
Viscosity at 20°C (cP)	2
Freezing Point	<-20°C
Self- Accelerating Decomposition Temperature (SADT)	≥ 55 °C
Purity as PAA (Peracetic Acid Content) (% w/w)	4.0 - 6.0
Hydrogen Peroxide Content (% w/w)	20.0 - 23.0

Microbiological Data

Test Method: Quantitative suspension test, DIN EN 1276 and EN 1650

Test Temperature: 20°C

Active ingredient: Peracetic Acid

Sterilization time [min]				
Test organisms	low organic load *			
concentration	0.025	0.1	0.5	1.0
Gram positive bacteria:				
Staphylococcus aureus ATCC 6538	5	< 5		< 5
Enterococcus hirae ATCC 10541	5	< 5		< 5



Prudent 5

Disinfectant with 5% Peracetic Acid

Gram negative bacteria:				
Pseudomonas aeruginosa ATCC 15442	5	< 5		< 5
Escherichia coli ATCC 10536	5	< 5		< 5
Yeasts:		15		< 15
Candida albicans ATCC 10231				
Moulds:				15
Aspergillus Niger ATCC 16404				

* 0.3 g/l Rinderserumalbumin;

Bactericidal efficacy

After examination of the bactericidal efficacy in the quantitative suspension test under low and high soiling, according to EN 1276, the efficacy (reduction ≥ 5 lg) on the test organisms *Staphylococcus aureus* ATCC 6538, *Pseudomonas aeruginosa* ATCC 15442, *Escherichia coli* ATCC 10536 and *Enterococcus hirae* ATCC 10541 can be proved. The bactericidal effectiveness of the product Prudent 5 was proved to be sufficiently effective under low soiling at a concentration of 0.025 % at 20° C within 5 minutes.

Yeasticidal efficacy

After examination of the yeasticidal efficacy in the quantitative suspension test under low soiling, according to EN 1650, the efficacy (reduction ≥ 4 lg) on the test organism *Candida albicans* ATCC 10231 can be confirmed. The yeasticidal effectiveness of the product Prudent 5 was proved to be sufficiently effective under low soiling at a concentration of 0.1 % at 20° C within 15 minutes.

Fungicidal efficacy

After examination of the fungicidal efficacy in the quantitative suspension test under low soiling, according to EN 1650, the efficacy (reduction ≥ 4 lg) on the test organisms *Candida albicans* ATCC 10231 and *Aspergillus Niger* ATCC 16404 can be confirmed. The fungicidal effectiveness of the product **Prudent 5** was proved to be sufficiently effective under low soiling at a concentration of 1 % at 20° C within 15 minutes. Use biocides safely. Always read the label and product information before use.



Prudent 5

Disinfectant with 5% Peracetic Acid

Make sure that there is adequate ventilation for concentrated chemicals. Always use appropriate chemical resistant gloves to protect your hands and skin and wear eye protection such as chemical goggles. Strictly avoid ingestion and inhalation.

Do not expose at high temperature. Store the product in original containers in cool and dry place, away from sources of heat, flame and direct sunlight. Very toxic to aquatic life with long-lasting effects. Please refer technical data sheet, label, and MSDS for more details and shelf life.

Safety Assessment

Eye contact: Flush with water for at least 15 minutes. If irritation occurs and persists, obtain medical attention.

Skin contact: Wash with plenty of soap and water. If irritation occurs and persists, obtain medical attention. Wash clothing before reuse.

Inhalation: Get fresh air. If difficulty in breathing or discomfort occurs, call a physician.

Ingestion: Drink one to two glasses of water. Do not induce vomiting. Do not give anything by mouth to an unconscious individual. Call a physician immediately.

Storage and Handling

All safety information on the handling and use of this product is provided in the Safety Data Sheet which can be consulted and / or obtained from our representatives

Keep this product in the original packaging or, where possible, in a tank of bulk reception, protected from extreme temperatures.

Declaration: Technical figures given in this data sheet are typical values based on internal laboratory measurements, following appropriate International Standard Methods, and should not be taken as a warranty or representation. It is the end user's responsibility to make sure that the product matches its requirements.

For more information on this product please contact your local Buzil Rossari representative.

Buzil Rossari Private Limited

402, OMEGA, Hiranandani Business Park, Powai, Mumbai – 400 076, Maharashtra, INDIA.

Tel.: +91-22-25701619 / 67411126, Mail: marketing@buzil-rossari.com, Web: www.buzil-rossari.com



Prudent 5

Disinfectant with 5% Peracetic Acid

Food Grade Quality Certificate

Application in food & beverage processing facilities

Here we confirm that, Prudent 5, if used according to the instructions in the product information sheet is suitable for the open plant cleaning in Beverage, Brewing, Dairy & Food Processing facilities and not to adulterate the food product in any manner during the use.

Essential condition is a thorough rinsing with clear portable water of surfaces intended to get in direct contact with foodstuff.

The remaining residuals that may contaminate or adulterate food contacting cleaned surfaces are expected to be negligible, following a potable water rinse. Therefore, the potential ingestion of residues resulting from the approved use of these types of products does not represent a significant health hazard after our assessment of the products as suitable for use under the intended conditions.



Thanking you,

Yours faithfully,

For Rossari Biotech Ltd.

Please note that this is not a regulatory assessment and should not be construed as an endorsement of regulatory compliance.



Analytical Test Procedure/Method

Product	Ross Prudent 5	Issued On	01/03/2018
Application	Disinfectant Liquid	Next Review On	01/03/2023

REAGENTS

0.1N KMnO₄, 0.1N Sodium Thiosulphate, 10N Sulphuric Acid, DM Water, Starch Indicator, 10% KI Solution

APPARATUS

Conical Flask, Burette, Burette Stand, Measuring Cylinder

PROCEDURE

- Take 50ml of test solution in which the concentration of Prudent 5 is to be determined, in a conical flask.
- Add 20ml of 10N Sulphuric Acid, mix well. Allow to stand for 3-4 mins.
- Titrate the solution in the conical flask against 0.1N KMnO₄, till the solution colour changes from colourless to pink. The pink colour should persist for a minimum of 15 sec.
- Add 10ml of KI solution, titrate with 0.1N Sodium Thiosulphate till a colour change of brown to yellow is obtained.
- Add 1-2ml of starch indicator.
- Continue the titration against 0.1N Sodium Thiosulphate, till the colour changes to colourless. Note the burette reading of Sodium Thiosulphate. Calculate the concentration of Prudent 5 as follows:

CALCULATION

Prudent 5 (%) = Burette Reading X 0.05

MATERIAL SAFETY DATA SHEET

according to REGULATION (EU) No 453/2010, annex I

Prudent 5

Date of issue: 08.06.2018

Page 1 of 7

1. Identification of the mixture and of the company / undertaking

1.1. Product identifier

Prudent 5

1.2. Relevant identified uses of the mixture and uses advised against

Disinfectant for professional use.

1.3. Details of the supplier of the safety data sheet

Manufacturer:

Manufactured under licence from buzil
ROSSARI BIOTECH LIMITED
Plot No. 10 & 11, Survey No.90/1/10 & 90/1/11/1,
Kumbharwadi, Village Naroli
Silvassa-396235
Dadra and Nagar Haveli (UT)
India

e-mail info@rossarimail.com

www.buzil-rossari.com & www.rossari.com

1.4. Emergency telephone number

+91-22-61233800/61233887

MATERIAL SAFETY DATA SHEET

according to REGULATION (EU) No 453/2010, annex I

Prudent 5

Date of issue: 08.06.2018

Page 2 of 7

2. Hazards identification



Risk phrases:

Hazards for man and environment

R 7 - May cause fire.

R 20/21/22 - Harmful by inhalation, in contact with skin and if swallowed.

R 35 - Causes severe burns.

Classification system : The classification complies with current EC regulations on dangerous substances and preparations

3. Composition / information on ingredients

Chemical characterization

Description : Mixture in water of non-dangerous ingredients and substances listed below.

Hazardous ingredients :

CAS : 7722-84-1 hydrogen peroxide C; Xn, O; R 5-8-20/22-35 15-30%

EINECS : 231-765-0

CAS : 64-19-7 acetic acid C; R 10-35 8.0-10.0%

EINECS : 200-580-7

CAS : 79-21-0 peracetic acid C, Xn; O, N; R 7-10-20/21/22-35-50 4.0-6.0%

EINECS : 201-186-8

Additional information : The full text of the relevant risk phrases is given in section 16.

4. First aid measures

General information : Remove contaminated clothing immediately.

Inhalation : Remove from source of exposure. Obtain medical attention if symptoms develop. In case of unconsciousness bring patient in to stable position for transport.

Skin contact : Wash skin immediately and thoroughly with water. Remove contaminated clothing. Obtain medical attention if symptoms develop.

MATERIAL SAFETY DATA SHEET

according to REGULATION (EU) No 453/2010, annex I

Prudent 5

Date of issue: 08.06.2018

Page 3 of 7

Eye contact : Wash immediately with copious amounts of water and obtain medical attention.

Ingestion : Remove material from mouth. Drink one or two glasses of water (or milk) and obtain immediate medical attention.

5. Firefighting measures

Suitable extinguishing media CO₂, extinguishing powder or water jet. Fight larger fires with water jet or alcohol resistant foam.

Protective equipment No special measures required

Additional information: Cool endangered containers with water spray jets.

6. Accidental release measures

Protection of personnel : Wear suitable protective clothing, gloves and eye / face protection.

Environmental precautions : Dilute with plenty of water

Method of disposal : Absorb onto dry sand or similar inert material. Dispose of the collected material according to the regulations.

7. Handling and storage

Handling (see also sections 8 and 15)

Information for safe handling : Ensure good ventilation/exhaustion at the workplace.

Information about protection against explosions and fires : No special measures required.

Storage :

Requirements for storage rooms and containers : According to local legislation.

Storage in one common storage facility : According to local legislation.

Further information about storage conditions : Keep container tightly sealed. Protect from heat and direct sunlight. Store in cool place.

8. Exposure controls / personal protection

Additional information about design of technical systems : No further data; see section 7.

Components with limit values that require monitoring at the workplace:

7722-84-1 hydrogen peroxide (15-30%)

MATERIAL SAFETY DATA SHEET

according to REGULATION (EU) No 453/2010, annex I

Prudent 5

Date of issue: 08.06.2018

Page 4 of 7

WEL (Great Britain) : Short-term value : 2.8 mg/m³, 2 ppm

Long-term value: 1.4 mg/m³, 1 ppm

64-19-7 acetic acid (5-15%)

IOELV (Europe) : Long-term value: 25 mg/m³, 10 ppm

Additional information : The lists that were valid during the compilation were used as basis.

Personal Protection equipment :

General protective and hygienic measures :

Keep away from foodstuffs, beverages and food.

Take off immediately all contaminated clothing.

Wash hands during breaks and at the end of work.

Avoid contact with the eyes and skin.

Breathing equipment : In case of higher levels or prolonged exposure use self contained breathing equipments.

Protection of hands : Tightly fitting safety glasses.

9. Physical and chemical properties

Information on basic physical and chemical properties

Appearance: Clear uncoloured liquid

Odour: Characteristic

pH-value(1%): <=2.0

Melting point / melting range: ND

Boiling point / boiling range: ND

Density (25 °C): 1.15 gm/cm³

Solubility in / miscibility with water: Miscible in water

Viscosity (25 °C): ND

Specific gravity: ND

10. Stability and reactivity

Thermal decomposition / conditions to be avoided : No decomposition if used according to specifications.

Dangerous reactions : Keep away from impurities, decomposition catalysts, alkalis, reducing agents and flammable substances.

MATERIAL SAFETY DATA SHEET

according to REGULATION (EU) No 453/2010, annex I

Prudent 5

Date of issue: 08.06.2018

Page 5 of 7

Hazardous decomposition products : Oxygen

11. Toxicological information

Acute toxicity :

LD50(oral) : Based on the toxicological classification the LD50 (oral) is estimated to be 200-2000 mg/kg. This

value is of no practical significance due to the corrosive nature of the product.

Primary irritant effect :

Skin contact : Harmful by absorption. Causes severe burns. Powerful oxidizing agent.

Eye contact : Causes severe or permanent damage.

Inhalation : Harmful if inhaled. Severe irritant, may cause pulmonary oedema.

Ingestion : Harmful. Causes severe burns. Ingestion will lead to a strong caustic effect on mouth and throat and to the danger of perforation of esophagus and stomach.

Additional toxicological information: NA

EC Classification: Harmful. Corrosive

12. Ecological information

General notes : Must not reach sewage water or drainage ditch undiluted or unneutralized. When used for its intended purpose, this product should not cause adverse effects in the environment.

13. Disposal considerations

Product :

Recommendation : Dispose off observing official regulations.

European waste catalogue :

16 09 03 peroxides, for example hydrogen peroxide.

Uncleaned packagings : **Recommendation** : Dispose off observing official regulations.

Recommended cleaning agent : Water, if necessary with cleaning agent.

MATERIAL SAFETY DATA SHEET

according to REGULATION (EU) No 453/2010, annex I

Prudent 5

Date of issue: 08.06.2018

Page 6 of 7

14. Transport information

Land transport ADR/RID (cross-border) :

ADR/RID Class : 5.2 Organic peroxides.

Kemler number : 539

UN-Number : 3109

Packaging Group : -

Label : 5.2+8

Proper Shipping Name: 3109 ORGANIC PEROXIDE TYPE F, LIQUID (PEROXYACETIC ACID)

Maritime transport IMDG :

IMDG Class : 5.2

UN-Number : 3109

Label : 5.2+8

Packaging Group : II

EMS Number : F-H, S-Q

Marine Pollutant : No

Proper Shipping Name: ORGANIC PEROXIDE TYPE F, LIQUID (PEROXYACETIC ACID)

Air transport ICAO-TI and IATA-DGR :

ICAO/IATA Class : 5.2

UN/IDNumber : 3109

Label : 5.2+8

Packaging Group : II

Proper Shipping Name: ORGANIC PEROXIDE TYPE F, LIQUID (PEROXYACETIC ACID)

Transport/ Additional information : Transport regulations include special provisions for certain classes of dangerous goods packed in limited quantities.

15. Regulatory information

Classification according to EC guidelines : The product has been classified and labeled in accordance with current EC Directives.

Symbol and hazard designation of product:



MATERIAL SAFETY DATA SHEET

according to REGULATION (EU) No 453/2010, annex I

Prudent 5

Date of issue: 08.06.2018

Page 7 of 7

16. Other information

The information in this document is based on our best present knowledge. However, it does not constitute a guarantee or any specific product features and does not establish a legally binding contract.